

# Flutter

Cross platform for building Mobile Apps

Summer Training • Academic Year 2025 – 2026

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# Course Agenda



06

## Session 6: Flutter Intro, Widget Tree & Basic UI

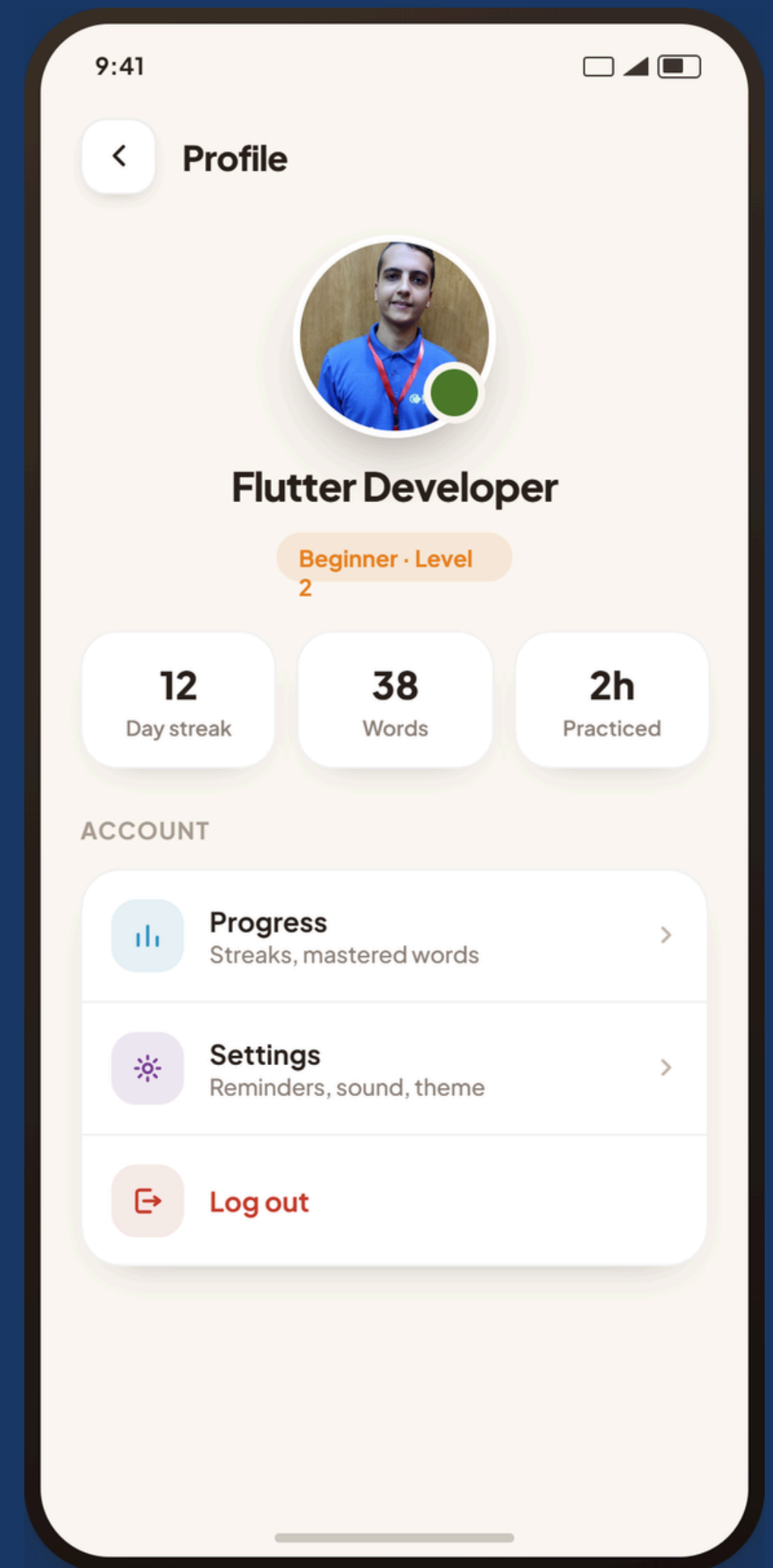
- install Flutter SDKs & Plugins, Emulator
- Intro to Flutter & UI/UX Role & Flutter Developer Role
- Architecture of Flutter & Widget Tree
- Stateless vs Stateful Widgets
- Run the first application in emulator
- Material App
- Basic UI Widgets (Scaffold, AppBar, Text, Icon, Container, Padding, Margin, Center, Image, Fonts)
- Layout Widgets (Row, Column, Stack, Positioned)
- Space Management (Expanded, Flexible, SizedBox)
- **Hands-on: Building a modern Language Learning App**
  - **Profile / Splash Screen**





# Splash Screen

# Profile Screen



# 01

# Ui / Ux designers role

"Before we write a single line of code, we need a blueprint. This is where the UI/UX designers come in.

- **UX (User Experience):** They study the users, figure out how the app should flow, and make sure it's easy to use.
- **UI (User Interface):** They choose the colors, typography, and design the actual screens using tools like Figma or Adobe XD. Basically, they give us the static pictures of how the app should look."





# 01

# Flutter Developer role

"Now, this is where WE step in. A design is just a picture; it doesn't do anything if you tap it. As Flutter Developers, our job is to bring these designs to life. We take the UI/UX screens and write the Dart code to build the actual application. The beauty of Flutter is that we write this code only once, and we generate apps for both Android and iOS. We build the screens, add the animations, and handle the user interactions."



# 01

## Backend Developer Role

"So, our app looks beautiful and the buttons work. But where does the data come from? How do we log in? This is the Backend Developer's job. They are the hidden brain of the application. They build the databases, handle the security, and create the APIs. As Flutter developers, we simply talk to the backend to send and receive data—like fetching a list of products or verifying a user's password





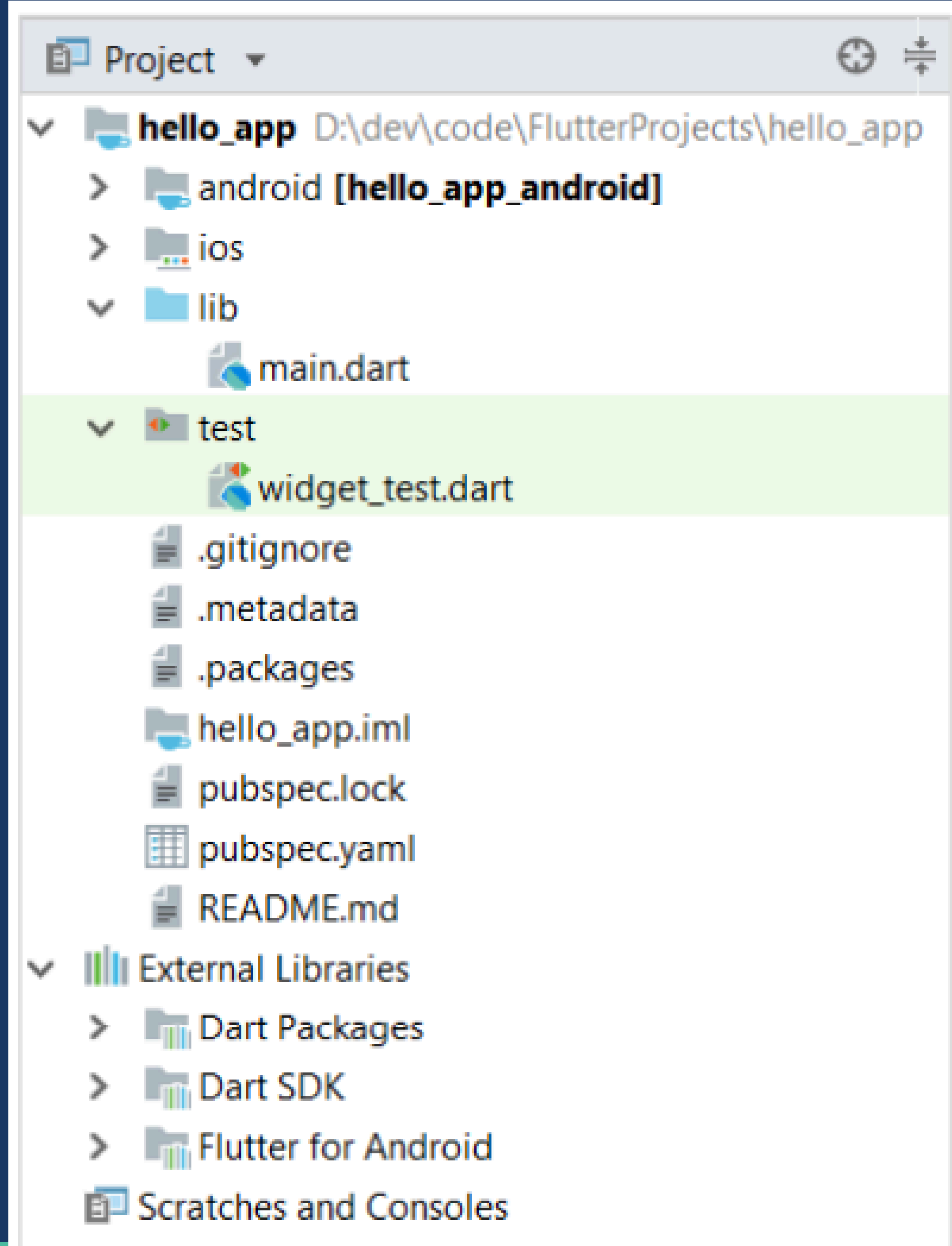
01

# Example



02

# Make your first Flutter App



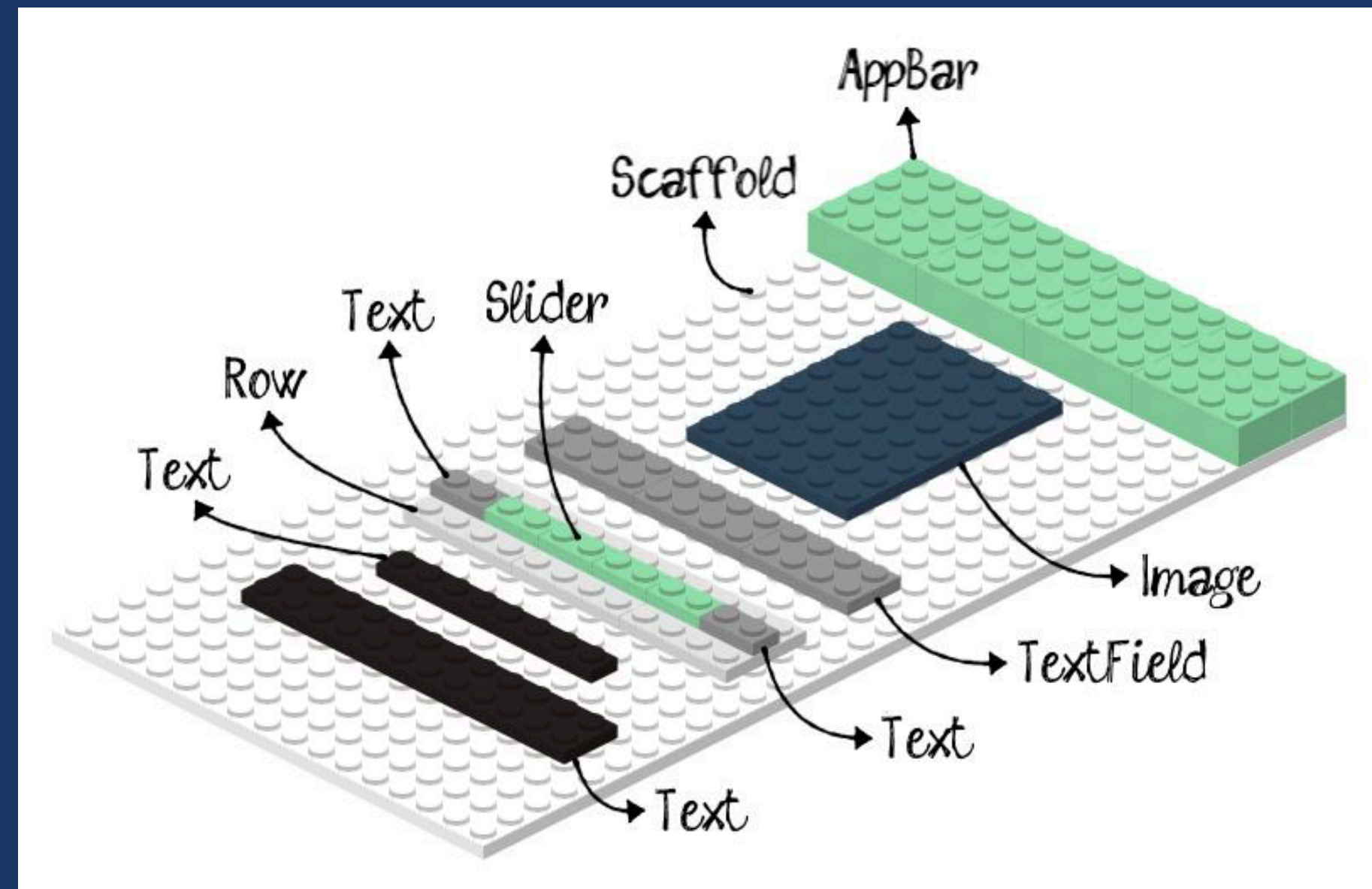


# 03

# Flutter Widgets Must knows!

## Everything is a Widget

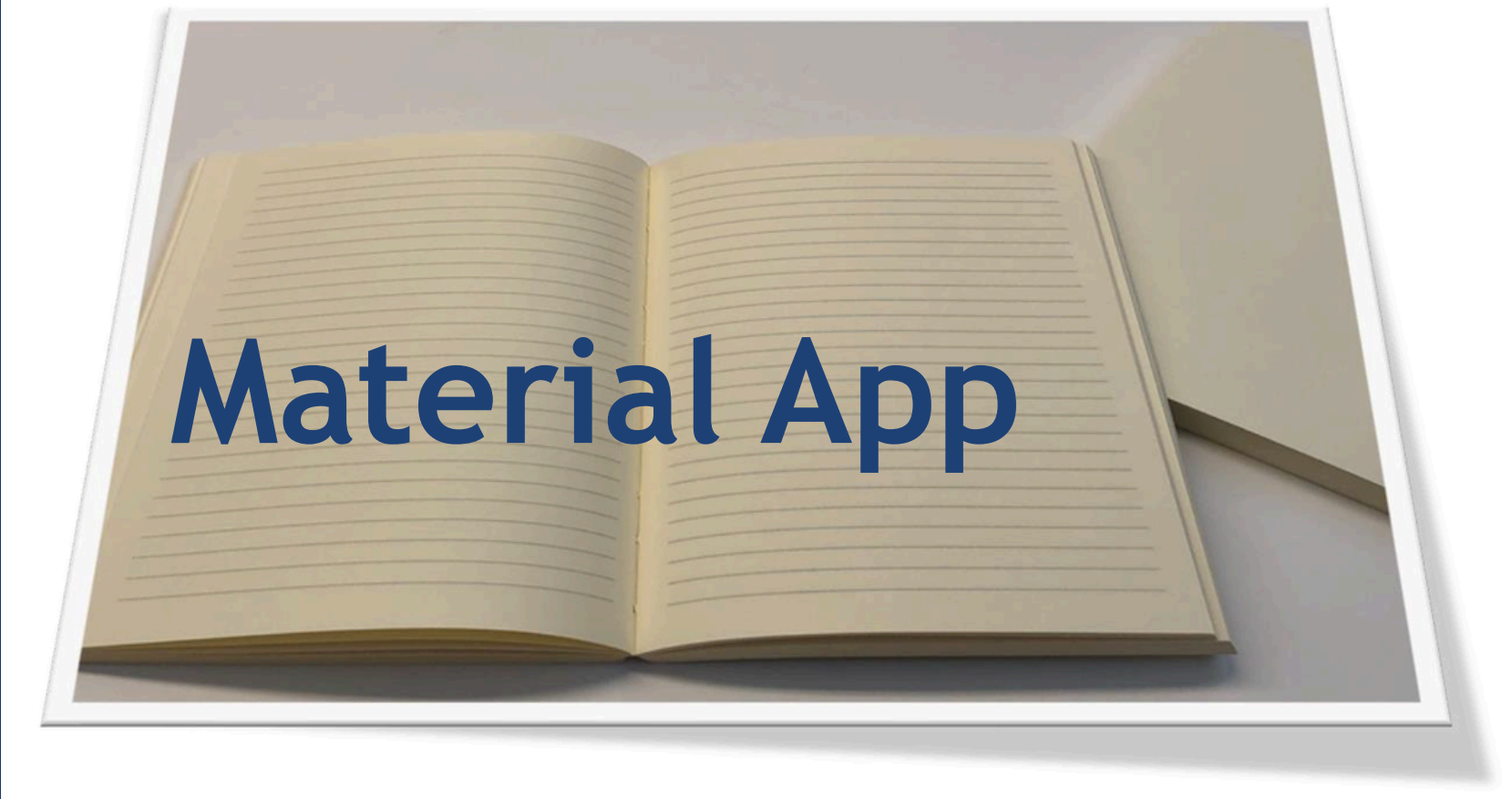
Widgets can be compared to LEGO blocks;  
By adding blocks together, you create an object,  
By adding different kinds of blocks, this alter the look and behavior of the object.  
Widgets are the building blocks of a Flutter app,  
Placing the widgets together creates the widget tree.



04

# Flutter Widgets Must knows!

## App/Page Setup Widgets



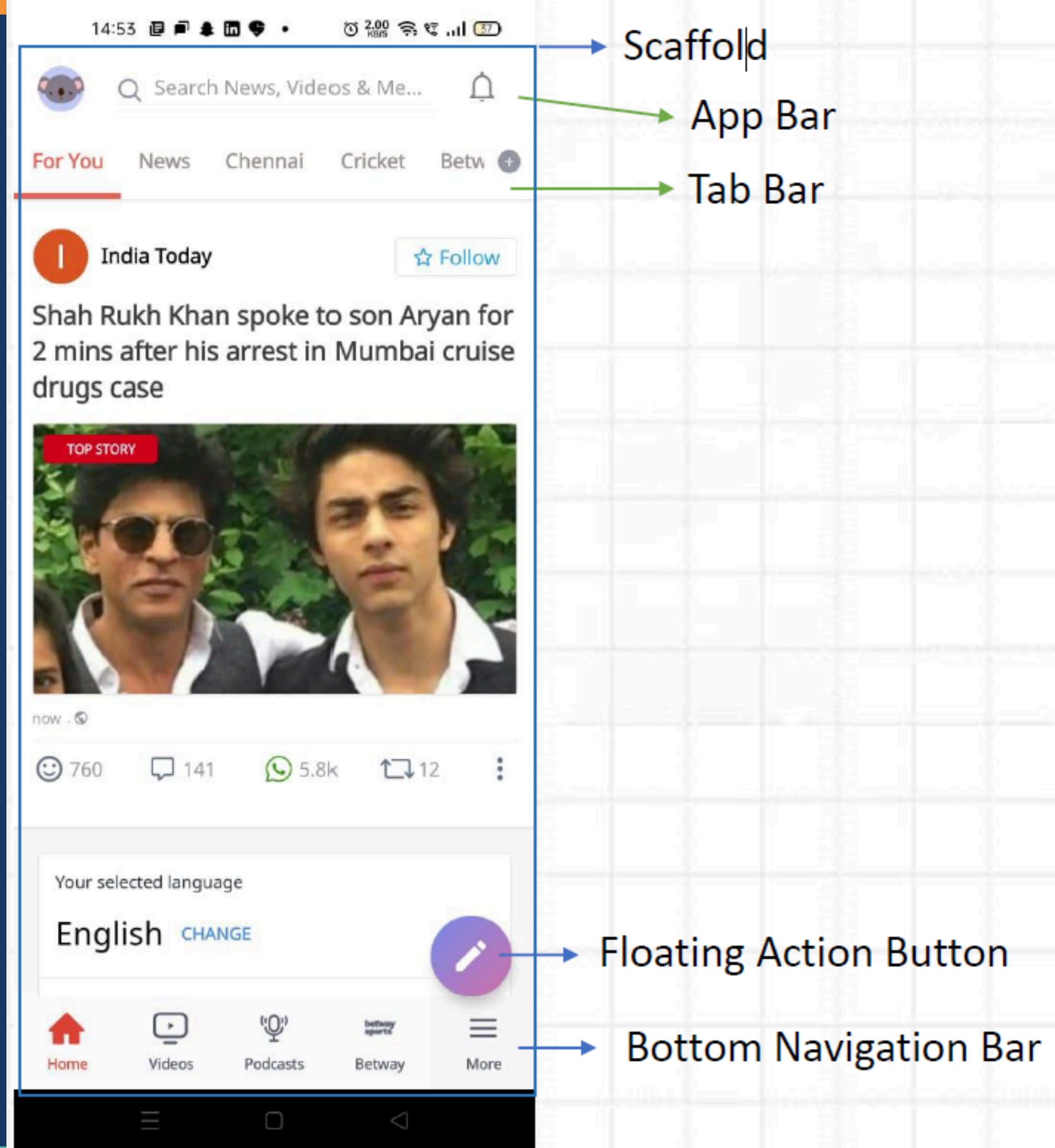
Material App



scaffold

04

# Scaffold (Page) elements widgets

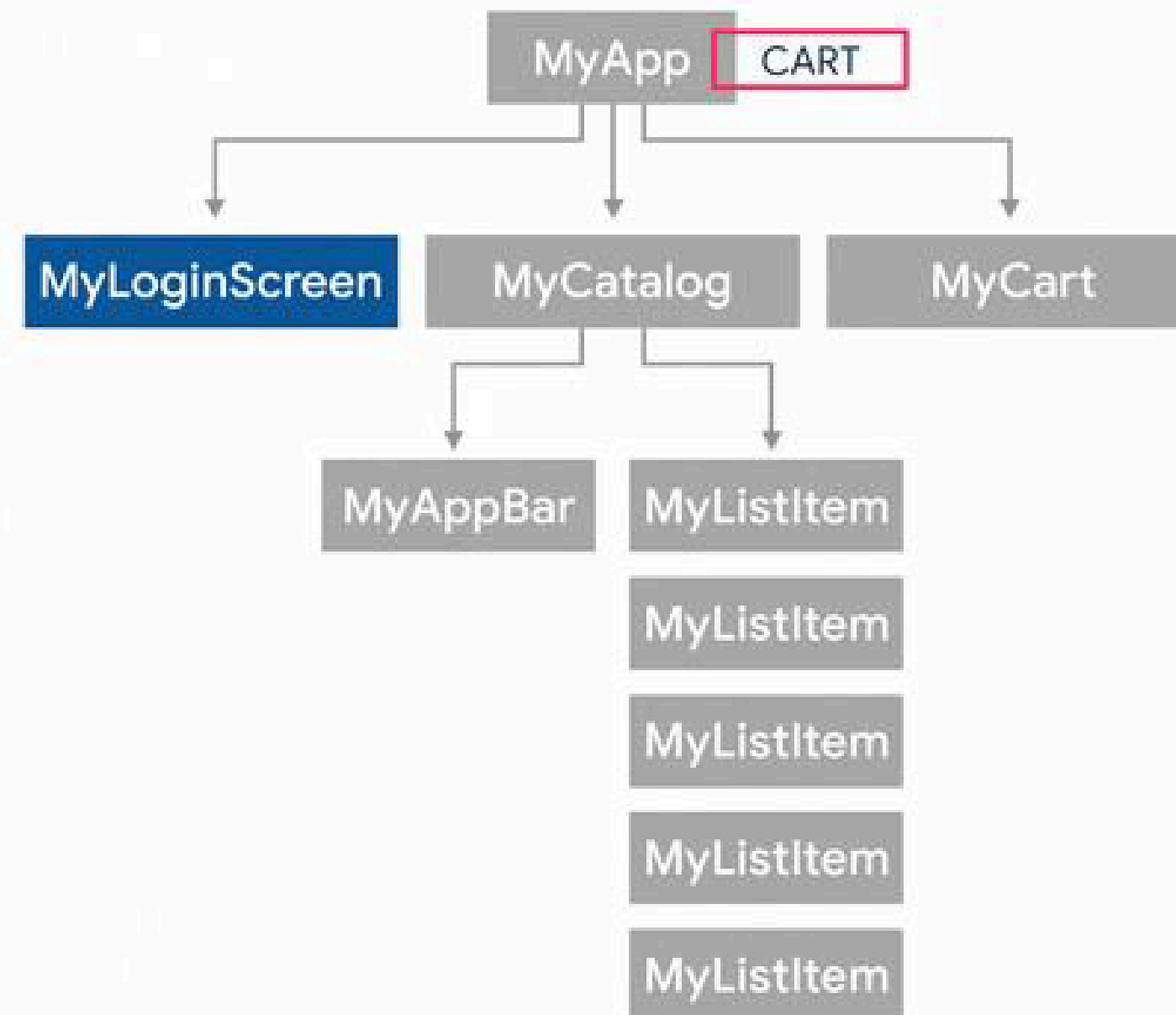
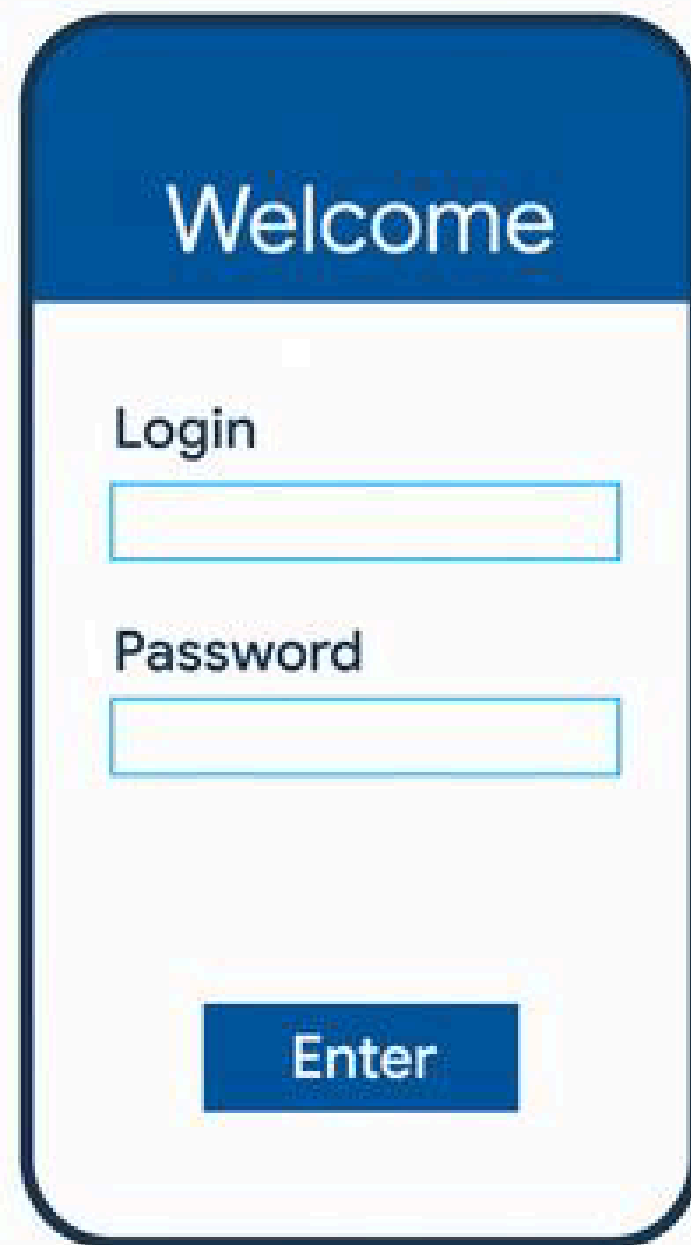




## 05

# Widget Tree

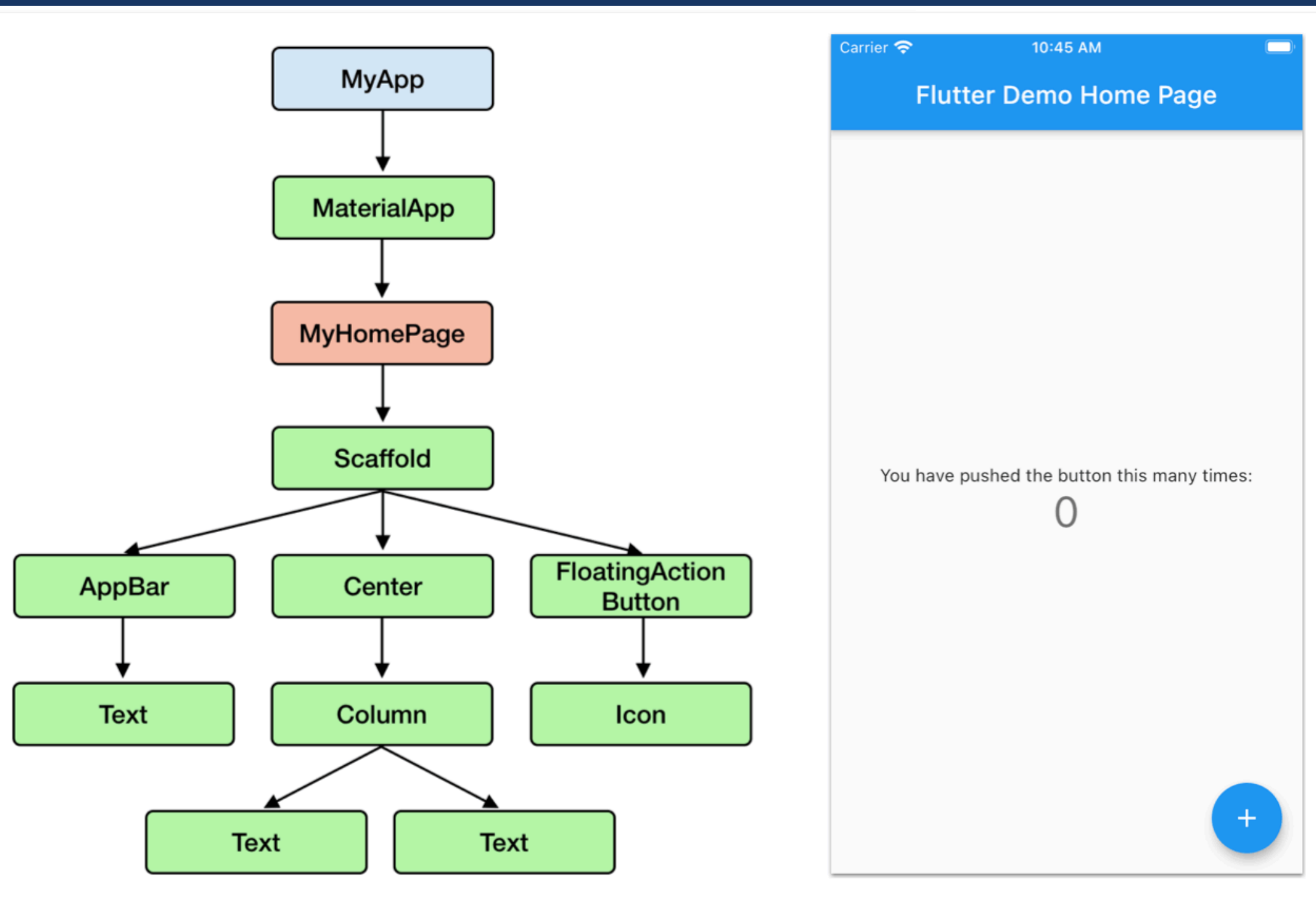
is a structure that represents how our widgets are organized.



# 05

# Widget Tree

is a structure that represents how our widgets are organized.



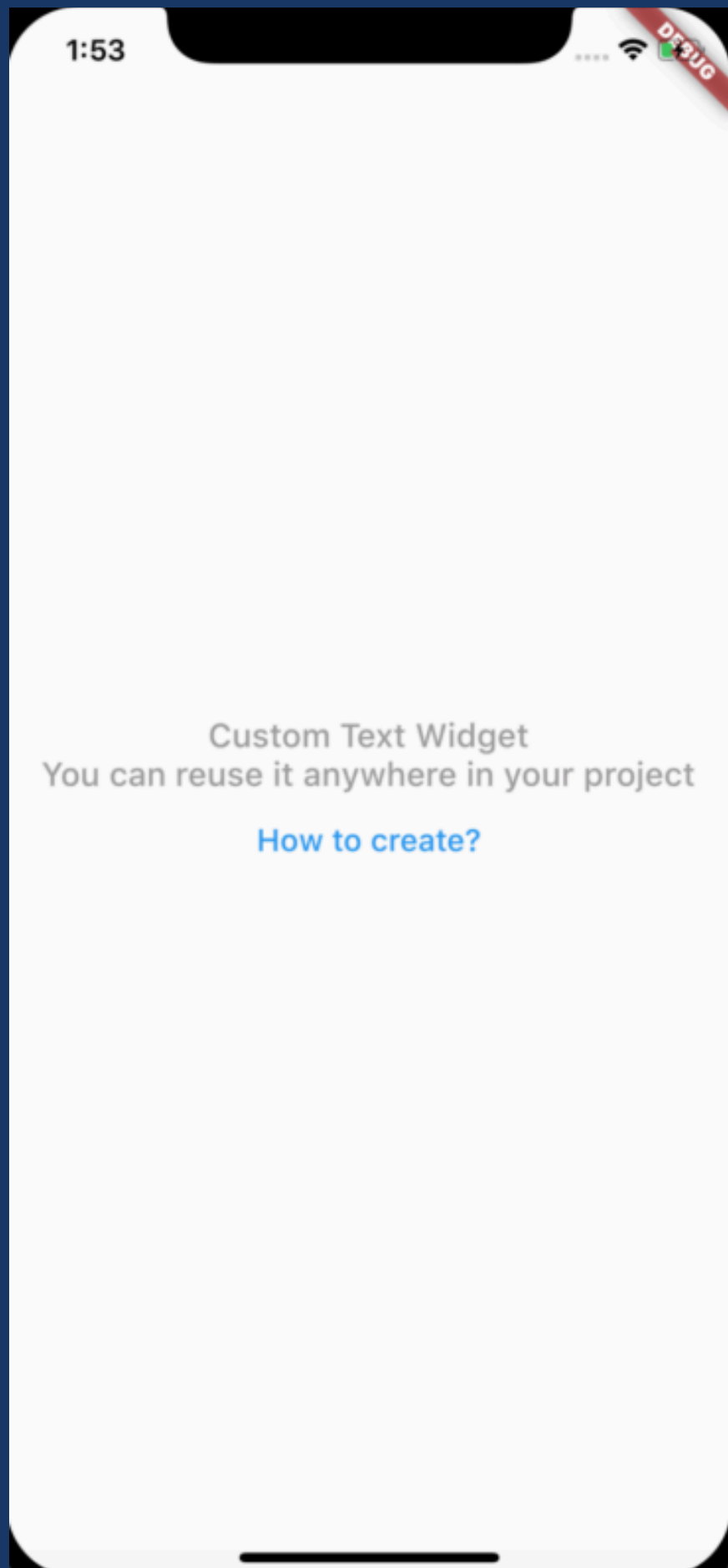
## 06

# Stateful and Stateless widgets

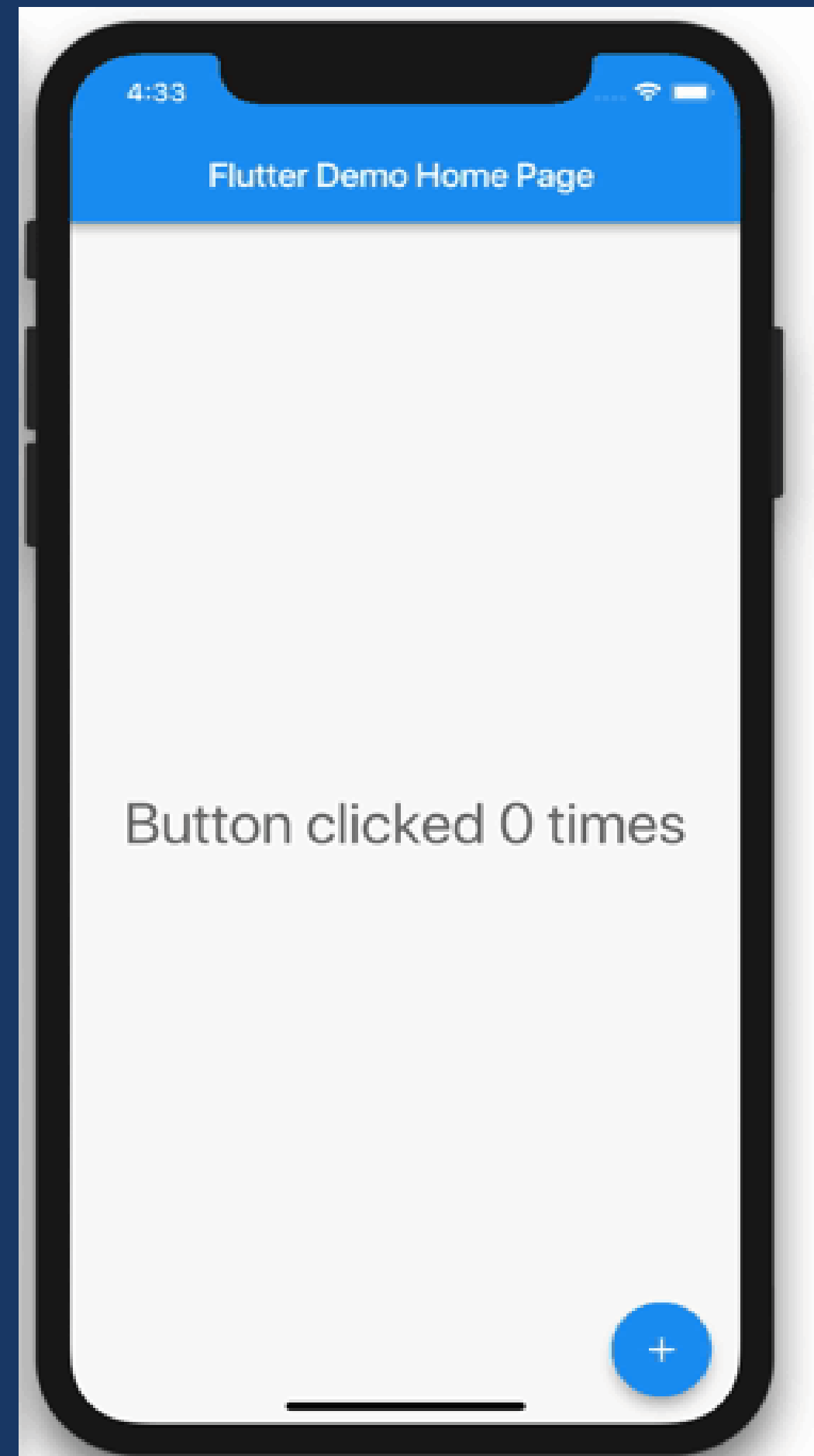
- ▶ If a widget can change appearance in response to events triggered by user interactions or when it receives data, it's **stateful**.
  - ▶ Checkbox, Radio, Slider, Form, and TextField.
- ▶ If a widget can't change, it's **stateless** widget.
  - ▶ Icon and IconButton.



06



Stateless widgets



Stateful widgets

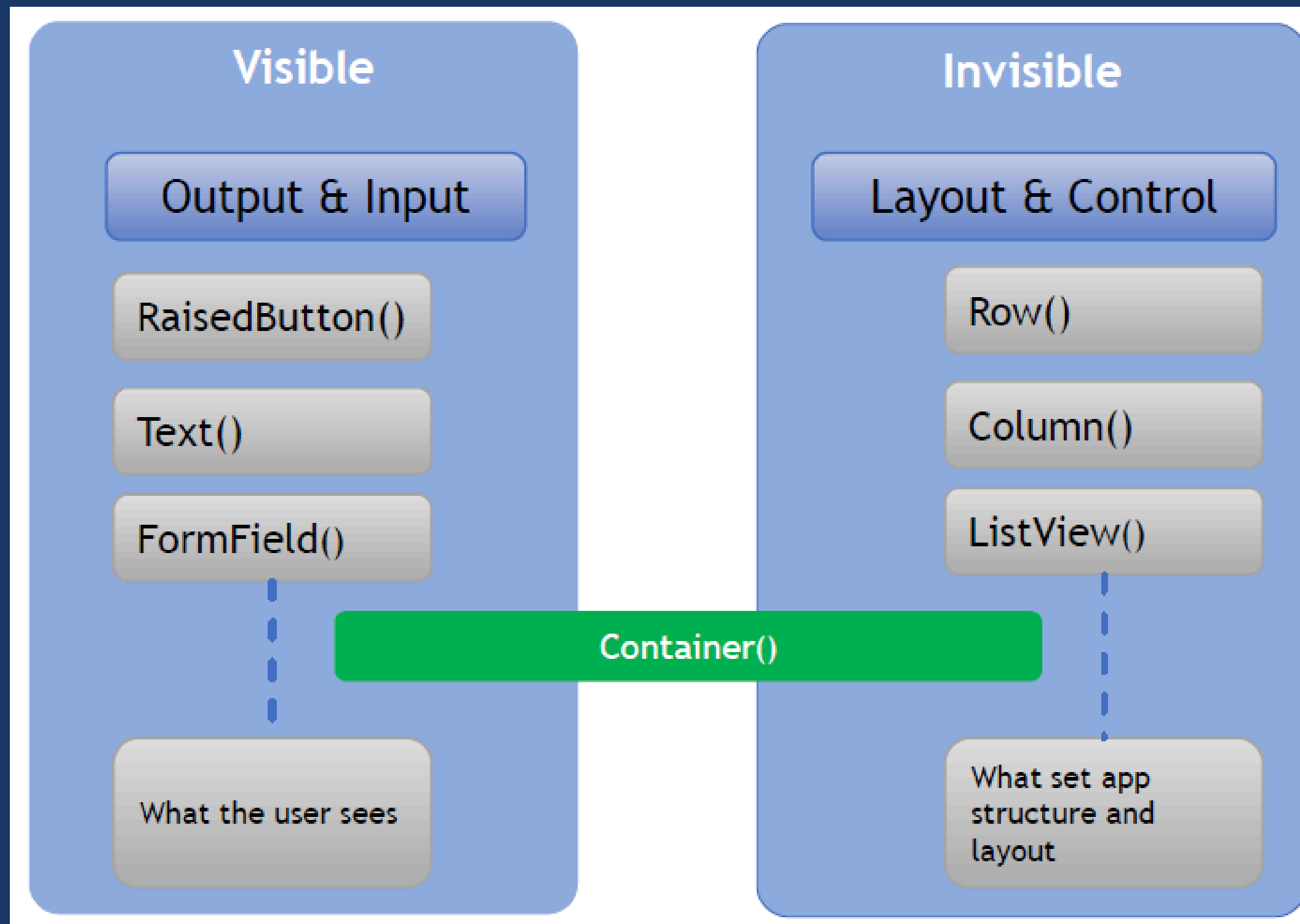
# 07

## Common widgets

- ▶ **Layout:** Scaffold, Column, Row, Stack.
- ▶ **Structures:** Button, Text, Image, RaisedButton.
- ▶ **Styles:** TextStyle, Color
- ▶ **Animations:** FadeInPhoto, transformations
- ▶ **Positioning and alignment:** Center, Padding

## 08

# Visible & Invisible Widgets





# 09

## User Interface: Material & Cupertino

- ▶ Google has its own user interface design called **Material**.
  - ▶ Material Design is a system for building beautiful digital products.
    - ▶ Style, branding, interaction, and motion under a consistent set of principles and component.
  - ▶ Most Flutter widgets support the Material design look and feel.
- ▶ Apple has its own user interface design based in **Cupertino**
  - ▶ Google has written many iOS-styled Widgets for Flutter developers



# THANK

*You*

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